

A PRODUCTIVE MARRIAGE

The New York State Energy Research and Development Authority (NYSERDA) found that PV systems on Energy Star homes make an impressive combination.

by **DIANNE GRIFFITHS**

As part of a demonstration program sponsored by the New York State Energy Research and Development Authority (NYSERDA) in 2002, home builders in the state were offered an incentive of as much as \$20,000 for the installation of a 2 kW PV system, provided that it was installed on an Energy Star model home in a subdivision of at least five Energy Star homes. The performance of the PV systems was to be monitored as part of the program. Yes, this was essentially a free PV system, and there were additional incentives for building to New York Energy Star Home (NYESH) standards. Sounds like an offer that's too good to refuse, but it took two years to find six builder projects that met the criteria and that were also geographically distributed around the state, which was an important consideration for the project.

There were many hurdles to face in convincing production builders to participate in the demonstration program, install PV systems, and deal with interconnection agreement issues. However, the performance of the installed PV systems on Energy Star energy-efficient homes in New York has been impressive. As of this writing, three of the monitored PV homes are occupied, and the annual solar fractions range from 30% to 50% of household electricity use. And the results from a home in the Bronx and a home in Syracuse provide an interesting contrast.

(For more on New York Energy Star Homes and the PV demonstration project, see "Pushing PV in New York.")



New York home builders were given an offer too good to refuse—practically free PV systems. This wintertime installation took place in Rochester.

Pushing PV and Efficiency

My employer, Steven Winter Associates, Incorporated, (SWA) and the National Association of Home Builders Research Center (NAHB-RC) were each awarded contracts by NYSERDA to implement this demonstration program. Our primary task was to recruit builders with qualifying projects and establish relationships among builders, raters, and PV system installers. Between the two contractors, SWA and NAHB-RC, 12 NYESH subdivisions of at least five homes each were to be established—two in each of six different utility service areas in New York State. At

least one home in each subdivision was to have a PV system.

The objectives of the demonstration project were as follows:

- Demonstrate NYESH at the community or subdivision level.
- Install residential PV systems to develop market interest, demonstrate their performance and reliability, and document the process of establishing interconnection agreements with utilities and code issues.
- Document the benefits of Energy Star and PV-equipped homes through homeowner interviews, monitoring, and utility bill analysis.

The following incentives were available to NYESH builders from the NYESH program:

- \$5,000 for a NYESH model home that is unoccupied and open to the public for a minimum of 60 days; and
- \$1,000 for each NYESH sold.

In addition to these standard NYESH program incentives, attractive PV system incentives were offered exclusively through this demonstration project. For a grid-tied PV system on a NYESH model home, 100% of the installed cost would be covered, up to the lesser of \$10 per AC watt installed or \$20,000. For the second and third PV systems to be installed in the same subdivision, the incentives were 75% and 60%, respectively, of the installed cost. At the time, market prices for PV panel systems were approximately \$9 per AC watt installed. Thus, the demonstration project systems, in most instances, were sized 2.4 kW–2.6 kW to receive the maximum incentive of \$20,000. In summary, a free 2 kW system was available for a NYESH model and second and third systems could be installed in the subdivision for \$5,000 and \$8,000,

respectively. The intent of the significant model home incentive was to have installed PV systems be available for public viewing and promotion.

The standard PV incentive available from NYSERDA when this demonstration project was initiated was \$4 per DC watt installed. Thus, the incentives available through this demonstration project were significantly higher for a modestly sized PV system. Currently, NYSERDA provides incentives of \$4.50 per DC watt installed for PV systems on NYESH homes. One builder that participated in the demonstration project has gone on to install larger PV systems on other homes and has applied for the standard incentive rather than the demonstration project incentive.

Finding Six Builder Participants

SWA tried several approaches to finding its share of candidate builders. These approaches included making cold calls, using lists of builders enrolled in Energy Star, and lists of local home builder association (HBA) members; and contacting the NYESH program regional account managers, PV installers, and HERS raters

for builder project leads. We used telephone calls and e-mail to make the initial contacts and held face-to-face meetings with all builders that expressed interest in learning more about the demonstration program. SWA and NAHB-RC coordinated frequently to avoid recruiting, and potentially confusing, the same builders.

SWA faced some challenges in recruiting builders for the demonstration program. The major challenges were as follows:

- There is a small speculative housing market in New York State. The few builders of larger subdivisions will construct a model home to be used as a sales office during the multiyear buildout of the subdivision.
- Uncertain and ever-changing construction schedules made it difficult to involve builders in the program. Timing was critical for finding projects at the appropriate stage, with all approvals in place and the model home yet to be built. In several instances, both SWA and NAHB-RC found and tracked candidate

Pushing PV in New York

The PV demonstration program began to take shape in late 2001, when the New York State Energy Research and Development Authority (NYSERDA) issued a request for proposals for “the implementation of a demonstration and awareness campaign, to inform builders, realtors, appraisers, bankers, consumers, and building code officials of the benefits of Energy Star homes and grid-connected photovoltaic (PV) systems for homes.” This initiative was a merger between two distinct programs within NYSERDA—Energy Star Homes and Residential PV.

Earlier in 2001, NYSERDA had launched the New York Energy Star Homes (NYESH) program. During the first six months of the program, more than 60 builders had become participants, over 35 homes were built to NYESH standards, and another 130 homes were committed. A large part of this early success was the result of a vibrant marketing and public

awareness campaign. The NYESH program’s “early adopters” were primarily custom home builders who built fewer than ten homes per year. A number of New York State’s largest production builders have since joined the program, and entire subdivisions of NYESH homes have been built. This expansion of the participating builder base has made NYESH homes available to a variety of new-home buyers.

The NYESH program has continued to grow steadily since 2001. The market penetration rate of New York Energy Star homes has increased from 0.29% in 2001 to over 13% in 2005. As of July 31, 2006, more than 7,800 homes have been built and labeled as NYESH.

Launched in 2000, the goals of NYSERDA’s PV programs are to develop and strengthen New York’s PV installation infrastructure, reduce the technical and market barriers to installing PV, and build market demand for PV. The New York

Energy Star Homes-PV demonstration program represented an opportunity for the PV program to leverage the marketing and greater public awareness of the NYESH program. The partnership between NYSERDA’s NYESH and PV programs has resulted in the installation of PV systems on over 40 NYESH homes.

In addition to the 40 PV systems installed on NYESH homes, 250 PV systems have been installed on existing homes. Owners of existing homes are encouraged to participate in New York’s Home Performance with Energy Star program in order to maximize the benefits of their PV system. As of last September, NYSERDA’s PV program was providing an incentive of \$4/W for PV systems installed on existing homes and \$4.50/W for PV systems installed on homes built through the NYESH program.

builder projects, but zoning approval or permitting issues delayed the projects so long that they could not be included in the program.

- The optimum lot for solar access may not be the most desirable lot for a sales model. Sales model homes are typically located on a prominent lot at the entrance to the subdivision. Such a lot may not be oriented for an unobstructed southern exposure.

- The New York State Electric and Gas service area is rural, with no major cities served and limited new construction. Ultimately, a grouping of homes by one builder on scattered lots was accepted in lieu of a subdivision.

- The aesthetic appearance of the PV panels was a concern to a few builders that were approached in the Hudson Valley area served by Central Hudson Gas & Electric.

Ultimately, over the course of two years, SWA successfully established six builder projects (see Table 1). (The projects are listed in the order in which the builders agreed to participate in the demonstration program.)

Overcoming Hurdles

Once a builder agreed to participate in the demonstration program, the next step was to facilitate the construction of the home with all of the necessary parties involved. Some projects went relatively smoothly, while others faced a few hurdles. These hurdles were related primarily to the installation of the PV system.

Timing and Responsibilities

In more than one instance, the completion of the electrical inspection was delayed because the electrical permit had been pulled without anyone indicating the presence of a PV system. This may have been because the decision to install a PV system came later in the process or, more likely, because the electrician overlooked the need to modify the permit application.

Two different PV contractors were used for the six builder projects. Both

Table 1. Six Builder Project Participants

Project	Location	Homes	HVAC	PV System*
Regional production builder's 18-lot third phase of a 63-home community	Syracuse	2-story, approx 2,000 ft ² plus full basement	Direct-vent gas furnace, 94 AFUE; ERV; 12-SEER A/C	2.4 kW AstroPower
Smaller builder with a 24-lot plot within a larger subdivision, plans to build out over several yrs	Rochester	2-story, approx 3,500 ft ² plus full basement	Direct-vent gas furnace, 92 AFUE; no A/C	2.4 kW AstroPower
40 3-family town houses, affordable housing	Bronx	3-story, 3,100 ft ² buildings, owner's unit and 2 apts in each building	Direct-vent gas boiler, 87 AFUE; hydronic baseboard; window A/C	2.2 kW AstroPower
3 adjacent duplex homes, affordable housing	Yonkers	2-story, 1,300 ft ² , 2 bedrooms, plus full basement	Direct-vent gas boiler, 84 AFUE; hydronic baseboard; window A/C	1.2 kW AstroPower
Regional production builder, 24-lot community w/2-acre zoning, 2-yr build out	Newburgh	2-story, 4,500 ft ² upscale	Direct-vent gas boiler, 93 AFUE; 12-SEER A/C	2.6 kW AstroPower
Log home manufacturer with small in-house construction group and 13-home community nearby	Catskills	1-story, homes range in size from 2,000 ft ² to 4,000 ft ² including a finished basement	Direct-vent propane boiler, 84 AFUE; radiant heat; no A/C	2.6 kW GE/AstroPower

* All of the PV systems were grid-tied. All of the inverters were Sunny Boy models by SMA.

requested that the electrical contractor play a role in the installation of the PV system. The electrical contractors were responsible for running the conduit from the PV panels to the inverter, installing disconnects, and connecting the inverter to the electrical panel. The cost of this work was included in the total project cost. In two instances, this arrangement caused delays because the electrical contractor did not do his work properly.

As the number of PV installations increases, electrical contractors will become more familiar with their part of the job. Until then, PV contractors should actively oversee this aspect of the installation. However, New York code requires that the electrical contractor be responsible for this aspect of the installation, and for the electrical inspection in new construction projects.

State Interconnection Requirements

In an attempt to simplify the process of installing grid-connected PV systems, New York State has established standardized interconnection requirements for small (2 MW or less) distributed gener-

ators that are connected in parallel with utility distribution systems. The requirements are listed in a 32-page document that details the application process, the design requirements, and the verification test protocols, and includes a 10-page agreement between the system owner and the utility. This document probably helped to make the work of installation easier, but in some ways it also helped to make it harder.

Within the standardized interconnection requirements is the recommendation that inverter equipment be selected from a certified equipment list maintained by the Public Service Commission (PSC). This language presented a challenge for one of the first PV systems to be installed in the demonstration program. The process of installing the PV system, from applying for an interconnection agreement to getting the PV system up and running, took five months to complete. The problems arose when a utility representative visited the site to conduct an inspection and an inconsistency was found between the installed inverter firmware and the inverter firmware on the certified equipment list. After several



The PV system installed on this new Syracuse home was monitored for 11 months. The system provided 50% of the household electricity needed during that time.

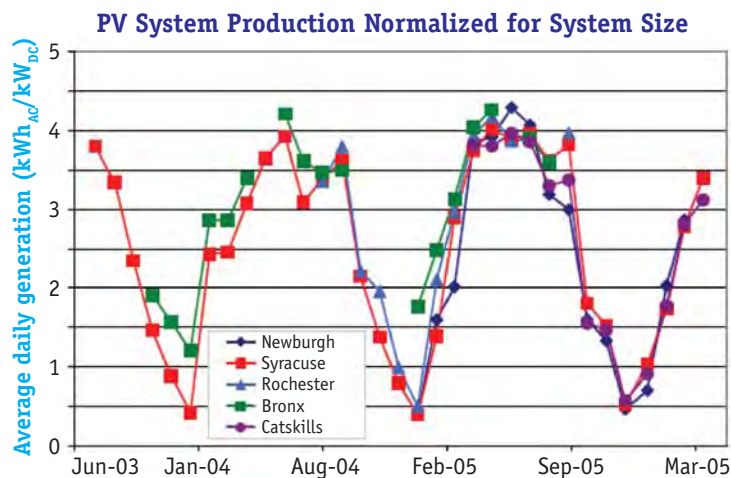


Figure 1. Each of the five operating PV systems has produced 2,300–2,600 kWh annually. This performance is in fairly good agreement with predicted estimates.

telephone and e-mail communications with the PSC and the inverter manufacturer, the firmware issue was resolved.

Insurance requirements posed a problem for another project. The utility told the builder that his insurance did not satisfy the interconnection agreement requirements and insisted on an insurance certificate from the homeowner. When this happens, the builder cannot install a PV system until the homeowner is identified. The Standard Interconnection Agreement clearly states

that general liability insurance coverage is not required to operate a PV system. However, the agreement does include the following statement:

Due to the risk of incurring damages, the Public Service Commission recommends that every distributed generation system customer protect itself with insurance, and requires insurance disclosure as part of this agreement.

The customer, or the owner of the PV system, must check off a box indicating

whether he or she does or does not have general liability insurance. Fortunately, no damages have been incurred on this project and therefore no insurance claims have been made. But one has to wonder how appropriate this language is if the builder is transferring the agreement to the buyer when the home sale closes.

PV Performance

After the PV systems were installed on the model homes, monitoring equipment was installed to record their performance over time. In five homes, DC and AC generation from the PV systems (and sometimes household electricity consumption) have been recorded at 15-minute intervals. In order to compare the systems effectively, we normalized these data by plotting energy generated (kWh_{AC}) per installed PV capacity (kW_{DC}) (see Figure 1). All of the systems are roof mounted, and all of them are oriented due south, except for the Newburgh system, which is oriented due west. Each of the five operating PV systems has produced 2,300–2,600 kWh annually. This performance is in fairly good agreement with predicted estimates.

Monitoring was performed at only five of the six builder projects because construction was completed on the affordable duplex homes in Yonkers too late in the project contract period to warrant monitoring. Construction began in 2003, but this project fell victim to the limits of volunteer services and a constant turnover of construction managers. Monitoring at the Syracuse home began in July 2003 and continued until March 2006. Monitoring at the Newburgh, Rochester, Bronx, and Catskills homes took place for various time periods between October 2003 and March 2006.

We compared the household average daily electricity use to the electricity produced by the home's PV system on a monthly basis for two of the occupied homes, one in South Bronx and one in Syracuse (see Figures 2 and 3). While PV system production for the two homes is comparable, the solar fractions are quite different. Natural gas serves the

space- and domestic water-heating loads for both of these homes.

The home in Syracuse became occupied in late May 2005. This four-bedroom 2,600 ft² home received a classic HERS score of 87, with a condensing gas furnace, a power-vented water heater, and central air conditioning. In September, the PV system met all of the household's electricity needs and sent 12 kWh of energy to the grid. For the total period monitored—11 occupied months—the PV system met 50% of the household's electricity needs.

The home in South Bronx became occupied in early 2004.

The building consists of a set of 30 town houses, and the owner occupies one of those town houses. The owner's unit includes the basement, the first floor, and half of the second floor. The town houses have hydronic baseboard heating served by a gas-fired boiler with an AFUE of 87. This boiler also serves the domestic hot water load via an indirect tank. The HERS score for the town house with the monitored PV system was 89. Window air conditioners were installed, and their summer time use is apparent in the graph (see Figure 3). Even so, the nonsummer electrical use for the home was approximately 15 kWh per day, significantly higher than that for the larger Syracuse home. The PV system provided approximately 30% of the Bronx home's annual electrical use. As Figure 3 shows, the PV system was disconnected during October and November. We believe that a security contractor turned off the PV breakers in the main panel when he was working on the home's security system. The homeowner was completely unaware of the problem until SWA told him about it during a visit to retrieve data from the logger.

A Successful Venture

Even though it involved only a few production builders and new construction projects, we feel that we gained several insights from conducting this program. Here are some of the most important things that we learned:

- The development and construction of residential subdivisions can sometimes take several years, from the time that the

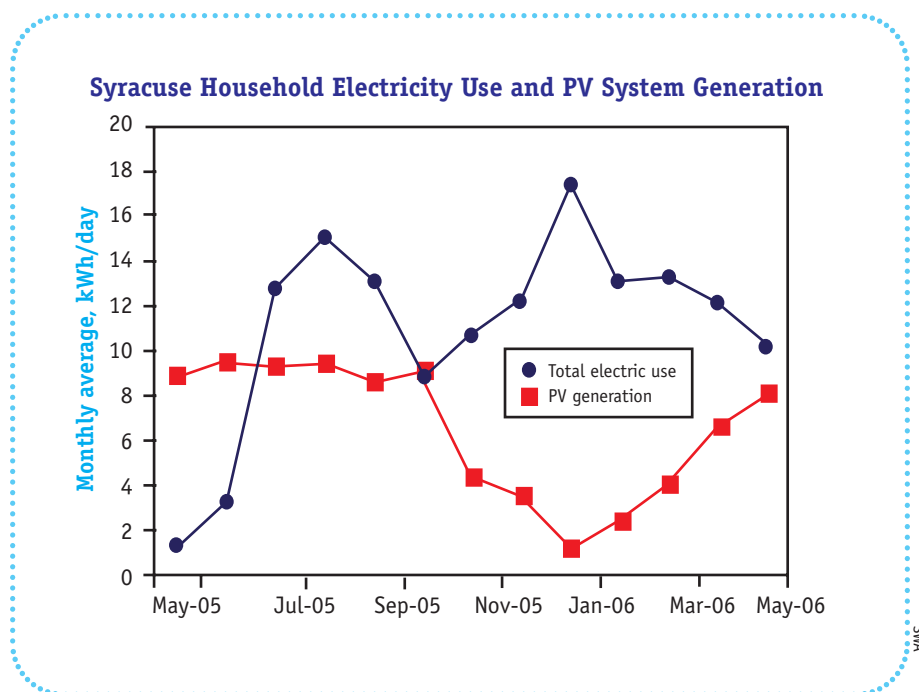


Figure 2. In September, the PV system met all of the household's electricity needs and sent 12 kWh of energy to the grid.

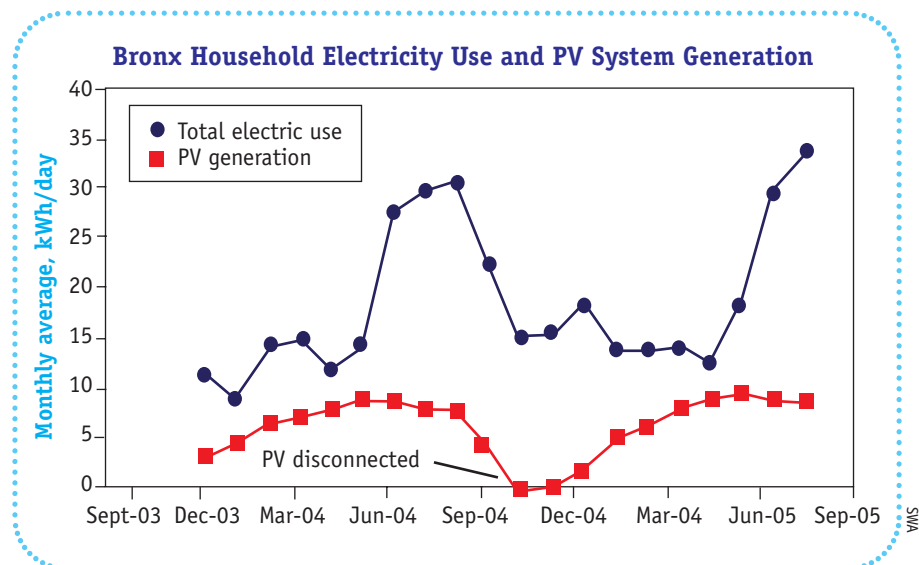


Figure 3. The nonsummer electrical use for the home was approximately 15 kWh per day, significantly higher than that for the larger Syracuse home (see Figure 2). The PV system provided approximately 30% of the Bronx home's annual electrical use.

builder applies for zoning approvals to the time that the home buyer moves in. Decisions on site planning that are critical to providing solar access for homes are made very early in the process. Incentive programs that can influence these decisions need to be long-term and stable, so that builders can make a multiyear commitment.

- Builders are unwilling to risk adding the cost of a PV system to a speculative home, for fear that potential

buyers will not recognize the value of the system. Few builders or realtors are knowledgeable enough, or are willing to take the time, to educate buyers on the value of a PV system.

- It can be difficult to establish an interconnection agreement between the utility and the builder that will be transferred to a home buyer within months.

- Because orientation and shading from trees and other structures vary from one site to the next, the performance,

and thus the appropriateness, of a PV system is not the same for all homes within a subdivision. A possible solution to this problem is to identify the PV-friendly lots on the site plan in the sales office. Buyers interested in the PV option would then select only from those lots. Electrical permit applications for homes on those lots would indicate an optional PV system.

- Selling speculative homes with PV systems to buyers who do not appreciate their value can result in unrealized benefits. This was the case for both of the affordable housing installations. The PV system had no influence on the buyer's decision to purchase the home, and thus the buyer doesn't care if the system is not functioning properly. This situation could also occur for a model home that is the last home to be sold in a desirable subdivision. A possible solution to both problems would be to establish marketing, rather than contractual, relationships between builders and PV contractors. The PV contractor would provide marketing materials for the builder to use in offering a PV

option. Any home buyers who showed an interest would be put in direct contact with the PV contractor. This arrangement allows the more knowledgeable and more highly motivated PV contractor to sell the concept, and the contract is then established directly between the home buyer and the PV contractor. The builder does little more than generate leads for the PV contractor, but builders can market themselves as environmentally friendly. This is the approach that is currently being taken by the log home manufacturer.

In short, we learned a lot that will allow the state of New York to continue to add reliable and renewable solar energy to its energy supply and to lesson energy use in the state's homes.



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FOR MORE INFORMATION:

To download the New York State Public Service Commission's 2005 report *New York State Standardized Interconnection Requirements and Application Process for New Distributed Generators 2 MW or Less Connected in Parallel with Utility Distribution Systems*, go to www.dps.state.ny.us/SIR_Require_11_04.pdf.

To download NYSEDA's 2005 report *New York Energy Smart Program Evaluation and Status Report*, Final Report, go to www.nyserda.org/Energy_Information/SBC/sbcmay05section6.pdf.